

Final Project Step 3

Final Project: Heart Attack Risk

By:

- Sofia Kruse Robertson 3004280
 - James Cha 9529140
 - Limeng Ye (7244478)
 - CHRIS ZARAGOZA (A1A0T32)
 - Zhuoqun Zhu (7257868)
-

Abstract and Introduction

Cardiovascular disease remains one of the leading causes of death worldwide, with heart attacks being a one of many concerns in public health in North America. As heart disease continues to affect millions annually, identifying and analyzing key risk factors is essential for prevention, early detection, and informed medical intervention. This project centers on the **Heart Attack Assessment Dataset**, a synthetic collection of 1,000 patient records created by **Shriyash Jagtap**. The data set, published on **Kaggle** on **January 29, 2025**, and retrieved on **April 24, 2025**, offers a valuable simulation of health data intended for use in statistical modeling.

The data set includes ten variables—six continuous numeric features (**age**, **total_cholesterol**, **ldl**, **hdl**, **systolic_bp**, and **diastolic_bp**) and four binary categorical variables (**sex**, **smoking**, **diabetes**, and **heart_attack**). These variables represent a combination of demographic, physiological, and lifestyle characteristics, enabling a comprehensive, multidimensional analysis of heart disease risk. The primary aim of this project is to determine which of these variables are strongly associated with the likelihood of a heart attack and to develop predictive models that can classify patients based on risk levels.

Guided by the hypothesis that elevated cholesterol, high blood pressure, smoking, and diabetes are significant contributors to cardiovascular events, we employ a wide array of data science

tools and modeling approaches. These include exploratory data analysis (EDA), logistic regression (both with and without interaction terms), t-tests, and adjusted linear regression models to investigate potential risk factors. We also evaluate predictive performance using cross-validation, confusion matrices, ROC curves, and AUC scores. Our models are implemented using `glm()` for generalized linear modeling, with packages such as `dplyr`, `ggplot2`, `e1071`, `ROCR`, and `car` used for data manipulation, visualization, and diagnostic analysis.

This project demonstrates how data science can be an informing healthcare tool in reinforcing preventative strategies against heart disease. The insights generated may be useful for guiding clinical screening and public health policy.

Citation:

Jagtap, Shriyash. Heart Attack Risk Assessment Dataset. Kaggle, 29 Jan. 2025. Retrieved 24 Apr. 2025, <https://www.kaggle.com/datasets/shriyashjagtap/heart-attack-risk-assessment-dataset/data>.

Question 1

Can risk be compensated with good metrics in certain variables? For example, can people with good metrics in one area (like high HDL) offset risks from risky behaviors like smoking?

Hypothesis 1 Exploration

Does high HDL offset the inherent risk carried by smoking?

$$H_0: \beta_{\text{HDL} \times \text{smoking}} = 0, H_1: \beta_{\text{HDL} \times \text{smoking}} < 0$$

Let us explore whether $\beta_{\text{HDL} \times \text{smoking}}$, or the interaction effect between HDL and smoking is 0 or negative.

Let us set up a basic equation.

$$\text{logit}(\text{Pr}(\text{heart_attack})) = \beta_0 + \beta_1 \cdot \text{Smoking} + \beta_2 \cdot \text{HDL} + \beta_3 \cdot (\text{Smoking} \times \text{HDL})$$

We will use the logit link function to test our theory; we use it since `heart_attack` is a binomial variable, and we want the probability to be between 0 and 1.

```
Call:
glm(formula = heart_attack ~ smoking + hdl + smoking:hdl, family = binomial(),
     data = heart_disease_sample)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.04422	1.06085	1.927	0.054 .
smoking	0.19896	1.59896	0.124	0.901
hdl	-0.10509	0.02484	-4.230	2.34e-05 ***
smoking:hdl	0.04051	0.03479	1.165	0.244

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 329.46 on 499 degrees of freedom
 Residual deviance: 269.06 on 496 degrees of freedom
 AIC: 277.06

Number of Fisher Scoring iterations: 6

```
Call:
glm(formula = heart_attack ~ smoking + hdl + sex + age + total_cholesterol +
     smoking:hdl, family = binomial(), data = heart_disease_sample)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-5.026947	2.012362	-2.498	0.012488 *
smoking	0.646052	1.659025	0.389	0.696968
hdl	-0.076843	0.026290	-2.923	0.003467 **
sex	0.983927	0.363774	2.705	0.006835 **
age	0.047027	0.012795	3.676	0.000237 ***
total_cholesterol	0.012232	0.004881	2.506	0.012208 *
smoking:hdl	0.037124	0.035916	1.034	0.301308

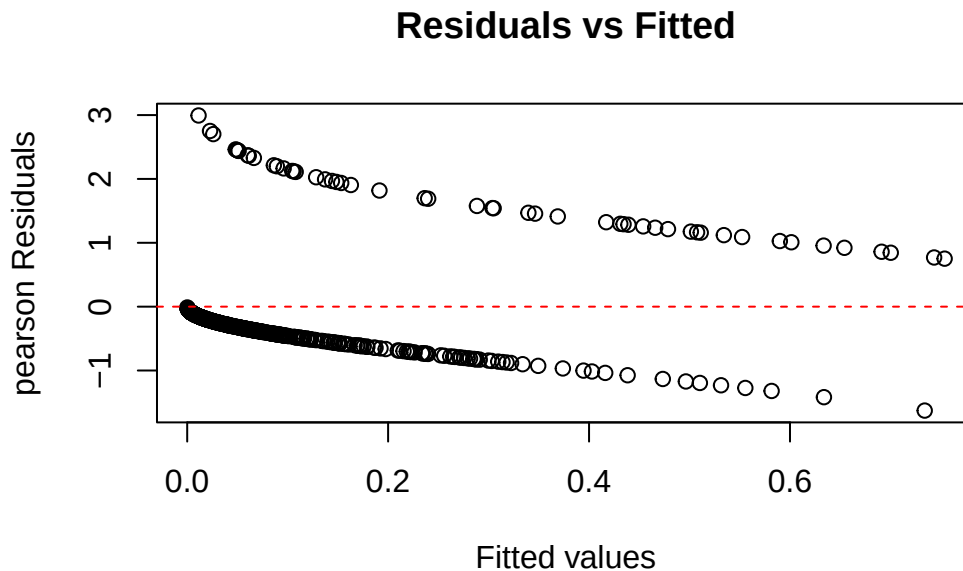
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 329.46 on 499 degrees of freedom

Residual deviance: 244.10 on 493 degrees of freedom
AIC: 258.1

Number of Fisher Scoring iterations: 6



The model seems to be under performing; with fitted values consistently going down as fitted values. Then again, this residual graph itself may not be a very reliable graph, as it does not seem to be designed for logistic regression.

<https://bookdown.org/jefftemplewebb/IS-6489/logistic-regression.html>

In either cases, HDL is protective against heart attacks overall, especially in non-smokers, with a negative estimate at -0.10509 (for base model) and -0.076843 (for full model) odds per mg/dL of HDL, with a p-value less than $\alpha = 0.05$ ($2.34e-05$ for the base model, 0.003467 for the full model).

In smokers, HDL's protective effect appears weaker, not stronger — and that difference is not statistically significant. Focusing on the full model, beta_2 (HDL coefficient for non-smokers) is at -0.0768, giving us the conclusion that each +1 mg/dL HDL multiplies the odds by $\exp(-0.0768) = 0.93$ (about -7%). $\text{Beta_1} + \text{beta_3}$ (HDL coefficient for smokers) is at $-0.0768 + 0.037124 = -0.039676$, giving us the conclusion that each +1 mg/dL HDL multiplies the odds by $\exp(-0.039676) = 0.96$ (about -4%). However, the difference between the two coefficients (Beta_2 vs $\text{Beta_1} + \text{Beta_3}$) is not significant, since the interaction term has a p-value of 0.30, which is over $\alpha = 0.05$; thus, we are unable to reject the null hypothesis that high HDL does not offset extra risk caused by smoking.

In short, HDL helps, but it doesn't "cancel out" smoking; quitting remains the safer choice. Based on both models and within this sampled dataset, there is no evidence that high HDL offsets smoking-related heart attack risk.

Hypothesis 2

Logistic

Here we split our data to training and testing data with a 80% split to training.

Table 1: Coefficients for Logistic Model with Interaction

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-24.500998	6.320875	-3.876203	0.0001061
diabetes1	13.189728	11.156504	1.182246	0.2371083
log(total_cholesterol)	4.116461	1.167477	3.525947	0.0004220
diabetes1:log(total_cholesterol)	-2.120109	2.103678	-1.007810	0.3135456

Table 2: Coefficients for Logistic Model with No Interaction

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-21.33834	5.2911788	-4.032814	0.0000551
diabetes1	1.95139	0.4617673	4.225916	0.0000238
log(total_cholesterol)	3.53132	0.9795010	3.605224	0.0003119

Here are our two models so far, one with an interaction, but we notice that the interaction is highly insignificant using the Bonferonni-adjusted significance level of $\frac{.05}{3} = 0.00333$. This suggests that the interaction between the presence of diabetes with a persons total cholesterol level, with a p-value of 0.3135456, does not give statistically significant evidence in having an effect on heart attack status. We also see that having the interaction changes the significance of having diabetes to a p-value of 0.2371083, whereas in our model without the interaction, it is significant at a Bonferonni-adjusted significance level of $\frac{.05}{2} = 0.0025$ with a p-value of 0.0000238. Otherwise, log(total_cholesterol) is significant in both models with a p-value of 0.0004220 in our interaction model and a p-value of 0.0003119 in our non-interaction model, suggesting the presence of higher cholesterol has statistically significant effect on heart attack status.

Table 3: AIC for Logistic Model with No Interaction

x
234.6611

Table 4: AIC for Logistic Model with Interaction

x
235.6923

We also take a look at our AIC scores, and take the lower and more significant model, which is the non-interaction model with an AIC score of 234.6610703, which is just barely an improvement. Here we have our coefficients for our chosen logistic model without the interaction $\log(\frac{p}{1-p}) = -21.33834 + 1.95139 \cdot \text{diabetes} + 3.53132 \cdot \log(\text{totalCholesterol})$ which gives us the log-odds of a heart attack. We use the equation $p = \frac{1}{1+e^{-21.33834+1.95139 \cdot \text{diabetes}+3.53132 \cdot \log(\text{totalCholesterol})}}$ for the probability of a heart attack occurring. The equation take the log of total_cholesterol.

Table 5: Chisq Test

	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
NULL	NA	NA	399	255.6439	NA
diabetes	1	11.76117	398	243.8828	0.0006048
log(total_cholesterol)	1	15.22171	397	228.6611	0.0000956

We see that our predictors add value to our model compared to the null model. The terms are added sequentially, first to last, and continue to decrease our residual deviance.

Table 6: VIF of Predictors

x
diabetes 1.077534
log(total_cholesterol) 1.077534

Our VIF for each predictor is less than 5 suggesting a lack of multicollinearity.

Table 7: Confusion Matrix for Training

	0	1
0	356	34
1	5	5

Table 8: Confusion Matrix for Test

	0	1
0	87	9
1	1	3

Table 9: Training Error Rate

x
0.0975

Table 10: Test Error Rate

x
0.1

Here we take a look at our confusion matrices of prediction accuracy and our training and test error rates, both of which look to be pretty low at 0.0975 and 0.1. We can compare this to our earlier model with the interaction. We have also tuned our decision threshold to .355 after slowly decreasing it from .5 to come up with the best testing and training error rates.

Table 11: Confusion Matrix for Training with Interaction

	0	1
0	355	36
1	6	3

Table 12: Confusion Matrix for Test with Interaction

	0	1
0	87	10
1	1	2

Table 13: Training Error Rate with Interaction

x
0.105

Table 14: Test Error Rate with Interaction

x
0.11

With the same tuning to the interaction model, we see a test and training error rate of 0.11 and 0.105, respectively which is just slightly worse and one of the coefficients is not statistically significant. We continue without the interaction.

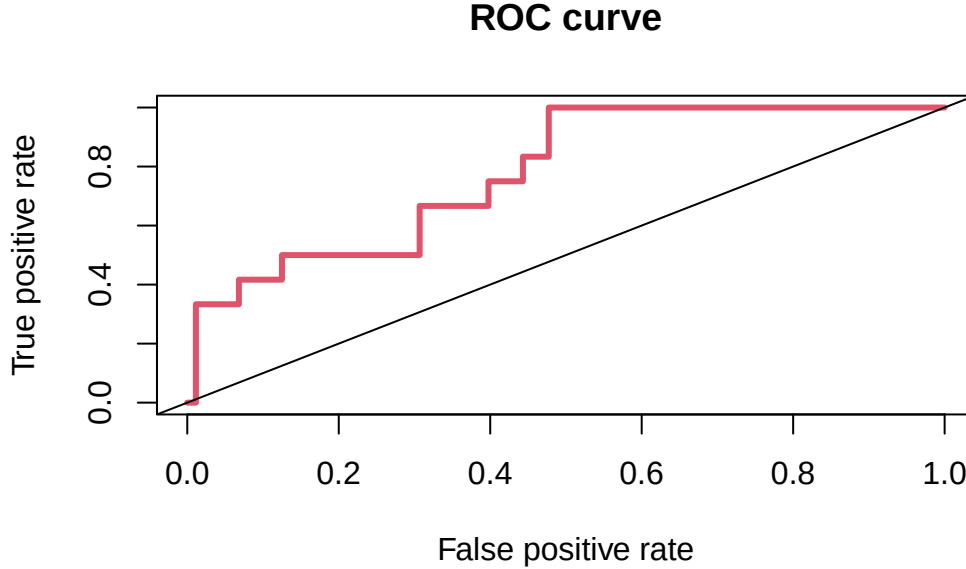
Table 15: Cross Validation for Test Error Rate on Logistic Model

fold	val.error
1	0.0981964
2	0.0981964
3	0.1002004
4	0.1022044
5	0.1002004
6	0.1022044
7	0.1022044
8	0.0981964
9	0.1022044
10	0.1002004

Here we apply Cross Validation with the same tuning to our chosen model without the interaction and validate the test error rate with an overall rate of 0.1004008.

Table 16: AUC for Logistic Model

x
0.7793561



We calculate an AUC of 0.77935606060606, which means that our logistic model is pretty accurate and has somewhat good reliability in its predictions.

Brief on Findings

Here we were able to somewhat accurately predict heart attacks using a logistic model with the predictors of having diabetes and the log of a persons total cholesterol level. We attempted to use the interaction between the two but found it insignificant, while having diabetes and the log of total cholesterol were found to be very significant in a model of their own. Going back to our original hypotheses, $H_0: \beta_1 = \beta_2 = \beta_1:\beta_2 = 0$, H_1 : at least one of $\beta_1, \beta_2, \beta_1:\beta_2 \neq 0$, this leads us to reject our null hypothesis of none of the coefficients having an influence on whether a person had a heart attack and find that there is statistically significant evidence that the presence of diabetes and the increase of the log of total cholesterol may increase the probability of occurrence of a heart attack.

Hypothesis 3 Exploration

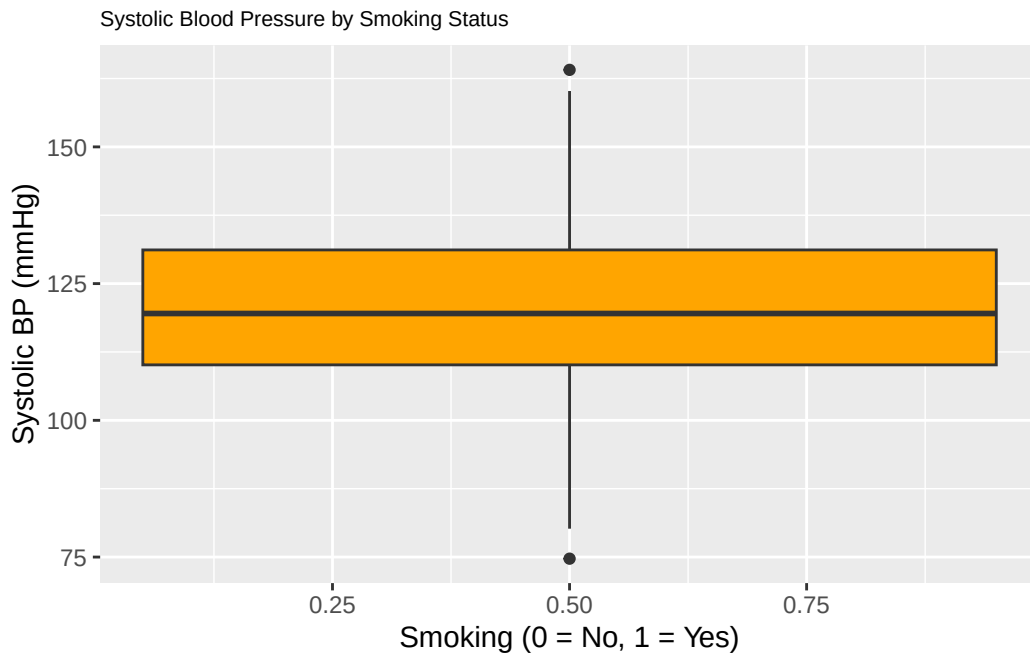
Smoking is widely known to contribute to cardiovascular problems, including elevated blood pressure. In this section, we are going to ask this question: Do smokers have higher systolic blood pressure?

H_0 : There is no difference in systolic blood pressure between smokers and nonsmokers. ($\mu_{\text{smokers}} = \mu_{\text{non-smokers}}$)

H_1 : Smokers have higher systolic blood pressure than nonsmokers. ($\mu_{\text{smokers}} > \mu_{\text{non-smokers}}$)

We want to find out whether smokers tend to have higher systolic blood pressure than non-smokers. Since systolic_bp is a continuous variable and smoking is binary, we can begin with a two-sample t-test, and a linear regression model.

Visualization with Boxplot



From the box plot, we find that:

Median Comparison: The boxplot shows that smokers tend to have a slightly lower median systolic blood pressure than non-smokers.

Two-Sample t-Test

Welch Two Sample t-test

```
data:  systolic_bp by smoking
t = 3.5373, df = 133.69, p-value = 0.9997
alternative hypothesis: true difference in means between group 0 and group 1 is less than 0
95 percent confidence interval:
    -Inf 9.540396
sample estimates:
mean in group 0 mean in group 1
    121.5240      115.0262
```

Since the p-value is far above 0.05, we fail to reject the null hypothesis that we do not have evidence to support the claim that smokers have higher systolic blood pressure.

Based on the t-test, we do not find statistically significant evidence that smokers have higher systolic blood pressure. The hypothesis is not supported in our sample, and additional modeling with control variables (like age) may be required to draw stronger conclusions.

Unadjust model

Call:

```
lm(formula = systolic_bp ~ smoking, data = heart_disease_sample)
```

Residuals:

Min	1Q	Median	3Q	Max
-41.089	-10.247	-1.149	10.161	42.557

Coefficients:

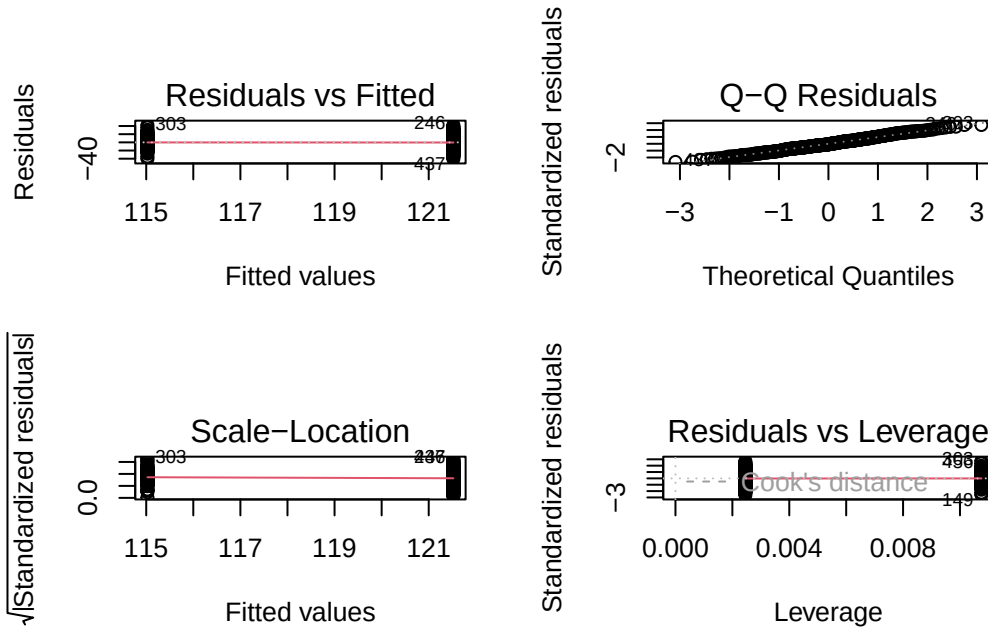
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	121.5240	0.7733	157.154	< 2e-16 ***
smoking	-6.4978	1.7930	-3.624	0.00032 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.6 on 498 degrees of freedom

Multiple R-squared: 0.02569, Adjusted R-squared: 0.02374

F-statistic: 13.13 on 1 and 498 DF, p-value: 0.0003199



Intercept ($\beta_0 = 121.52$): Indicates that the average systolic blood pressure (SBP) for non-smokers is 121.52 mmHg

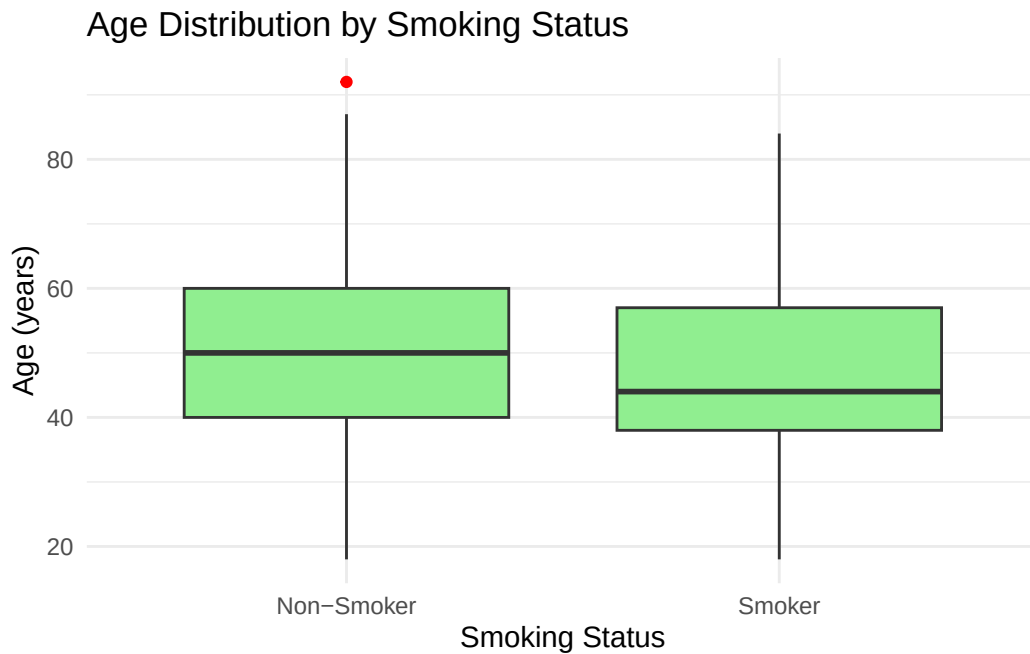
Smoking coefficient ($\beta_1 = -6.498$), ($p = 0.00032 < 0.05$): Indicates that smokers have on average 6.498 mmHg lower SBP than non-smokers, and this difference is statistically significant.

R^2 0.0257: Suggests that smoking status alone explains only 2.6% of the variation in systolic blood pressure. The remaining 97.4% is likely due to other factors such as age, body weight, heredity, or medication.

Diagnostic plots (residuals vs fitted, normal Q-Q, scale-location, residuals vs leverage) show no clear violation of model assumptions, suggesting that the linear model is reasonably appropriate.

In conclusion, without adjusting for any confounders, the data show that smokers tend to have significantly lower systolic blood pressure than non-smokers, in the opposite direction of the original hypothesis.

Checking for confounders



Welch Two Sample t-test

data: age by smoking

t = 2.4505, df = 139.02, p-value = 0.01551

alternative hypothesis: true difference in means between group 0 and group 1 is not equal to

95 percent confidence interval:

0.7563491 7.0756765

sample estimates:

mean in group 0 mean in group 1

50.37838 46.46237

Mean age of non-smokers: 50.38 years

Mean age of smokers: 46.46 years

Since the p-value smaller than 0,05, we conclude that smokers are significantly younger than non-smokers in this sample.

Because age is known to correlate with systolic blood pressure, this indicates that age is a likely confounder: younger individuals tend to have lower blood pressure.

Adjust Linear Regression

Call:

```
lm(formula = systolic_bp ~ smoking + age + sex, data = heart_disease_sample)
```

Residuals:

Min	1Q	Median	3Q	Max
-48.956	-10.060	-0.228	10.027	42.951

Coefficients:

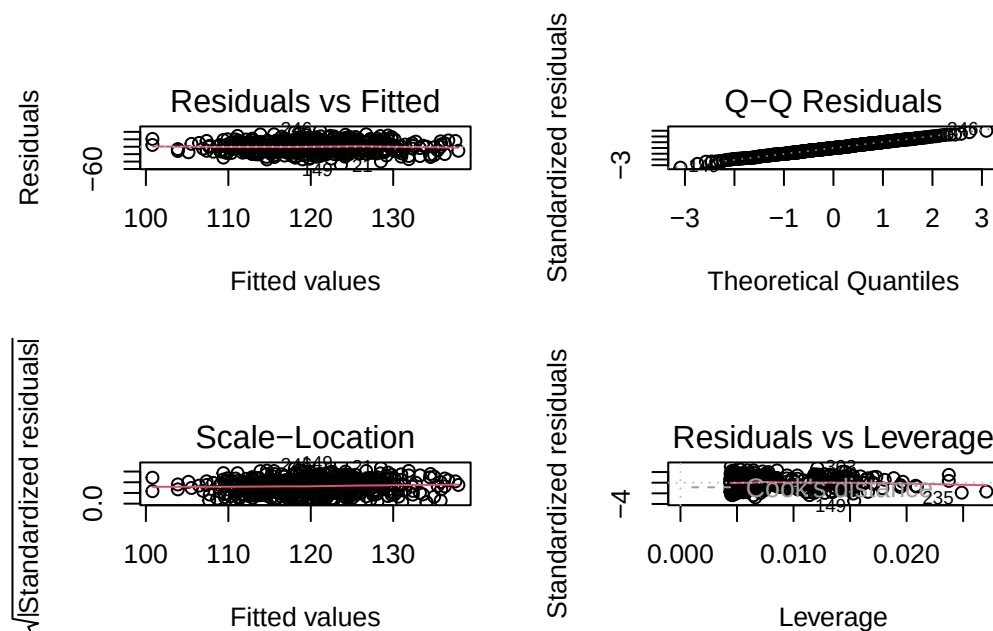
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	98.33302	2.58503	38.039	< 2e-16 ***
smoking	-5.27950	1.67420	-3.153	0.00171 **
age	0.43014	0.04618	9.314	< 2e-16 ***
sex1	3.08064	1.30691	2.357	0.01880 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.4 on 496 degrees of freedom

Multiple R-squared: 0.1731, Adjusted R-squared: 0.1681

F-statistic: 34.62 on 3 and 496 DF, p-value: < 2.2e-16



Intercept (98.33): This is the predicted systolic blood pressure for a baseline individual—a zero-year-old female non-smoker.

Smoking (−5.2795, $p = 0.00171$): After controlling for age and sex, smokers have on average about **5.28 mmHg lower** systolic blood pressure than non-smokers. Because the p-value is 0.00171 (< 0.05), this difference is statistically significant. In other words, once we account for age and sex, smoking status is associated with a lower predicted systolic BP in this sample.

Age (+0.43014, $p < 2 \times 10^{-16}$): Each additional year of age is associated with a **0.43 mmHg increase** in systolic blood pressure, holding smoking status and sex constant. This effect is highly significant, confirming that blood pressure rises steadily with age.

Sex (+3.08064, $p = 0.01880$): Males ($\text{sex} = 1$) have an average of **3.08 mmHg higher** systolic blood pressure than females, controlling for smoking and age. Since the p-value (0.01880) is below 0.05, this sex difference is also statistically significant.

Model Fit ($R^2 = 0.1731$, Adjusted $R^2 = 0.1681$): The three predictors together explain about **17.3%** of the variance in systolic blood pressure. In other words, age, sex, and smoking status account for a modest but meaningful portion of the differences in systolic BP across individuals in our sample.

conclusion

Although our original hypothesis predicted that smokers would exhibit higher systolic blood pressure, the adjusted regression reveals the opposite once age and sex are taken into account: smokers actually have, on average, lower systolic BP than non-smokers—about 5.28 mmHg lower—and this difference is statistically significant. Age remains the strongest predictor (roughly +0.43 mmHg per year), and being male adds about +3.08 mmHg. The fact that smoking status “reverses” when age is controlled indicates that smokers in our sample are generally younger, which masks the true association. In short, the observed lower blood pressure among smokers is explained by their younger average age; after adjustment, smoking itself does not raise systolic blood pressure.